

Lec 10 Introduction to Autoencoder

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Lecture	10 / 63 + intro
Duration	34:48
Captions	English (auto-generated)
Words	5,192
Video	aBauhB6K0jg
Playlist	Fundamentals of Generative AI and LLMs

Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI large language models. We are in the week two. We'll be starting with the auto encoders. In the previous session, we had seen the introduction to generative AI. We have also seen some of the fundamental concepts that are required from deep learning such as you know auto we we had seen activation functions, loss functions, then the different optimizers that are used. So with all those fundamentals and we have also seen the architecture of CNN, the working mechanism of CNN.

[00:38] In this particular week, we'll be starting with the auto encoders which we call it as in representation learning model. The following are the contents for the week two. We will first start with introduction to auto encoders, then we will move to an abstract overview of the architecture of auto encoders. Then we will see in detail how exactly the auto encoders works. » [snorts] » Then there are two different types when it comes to a latent space. One is under complete, second is over complete. We'll be looking into that. Then what is the loss function that we'll be using in auto encoders. We'll be using reconstruction loss. You'll understand why it is reconstruction loss.

[01:21] Then we will move towards the different types of auto encoders. Primarily, we'll be focusing on simple, deep, and then CNN based auto encoders. Then what

are the different regularization techniques that are there in auto encoders such as denoising, sparse, and then contractive auto encoders. And towards the end, we will solve one numerical example so that it will be a clear understanding to the learners how exactly the auto encoders works. And before we move into the hands-on, we will see what are the limitations of auto encoders. With this limitation, we will be understanding why there is a need for variational auto encoder.

[02:03] Towards the end of this week, we will see an hands-on implementation on different types of auto encoders. With this, let us start with the today's session. And in today's session, since we are starting with week two, we will see what is the introduction, why we require auto encoders, and then we will move towards the architecture. And [snorts] towards the end of this particular session, we will see under complete and overcomplete auto encoders.

[02:33] Okay. Where are we coming from? So, so far we have seen the neural network architectures such as ANN, CNN, RNN. So, we have seen these deep learning models. Whereas, all these are primarily used for supervised learning. Now, what is a supervised learning? If you see, this is one of the example for supervised learning, where we have a samples, and we have a label associated for each of these sample. So, therefore, a supervised learning will have an input X and then the label Y .

[03:08] » [snorts] » With this, what exactly the model will focus on all of these models? It mainly focuses on learning the mapping of from input to output. So, it learns the mapping of for this particular input, what is the label which is associated. So, this is all about the supervised learning, where we have an input data, and we have a labels associated with it. And the model is mainly aiming to learn the mapping between X and then the Y .

[03:39] » [snorts] » But, what when we don't have a labels with us? So, for example, if you consider this, here it is a data set which is having an age, height, and then the weight. So, here we don't have any of the labels. Then, what kind of model learning a problem it is? This is an unsupervised learning. In unsupervised learning, what is that the model is mainly aiming? It is a directly learning the data. Um the models learns directly from an unlabeled data here. We don't have any of the labels. So, this is a not there.

[04:15] Label is not there. So, whatever the model is learning, the model is learning directly from this particular data. So, this is what is the difference between supervised and unsupervised. In supervised, we have labels also. And the mainly, the model is aiming to learn this mapping. Whereas in unsupervised, we don't have the labels. When we don't have the labels, the model is directly learning the learning from this data.

[04:43] Now, all what we have in unsupervised is we just have the We have an input to the data. And here, the input data is represented as X . And what is that the

model is trying to learn? It learns what is a hidden pattern. Is there any correlation between the features? And what exactly is the structure of the data? This is what is the primary objective of an unsupervised learning models. So, while unsupervised learning broadly focuses on discovering the patterns or the structure in unlabeled data, there is a special category of unsupervised learning models. And what this a special category primarily aims? To learn the meaningful representation of the data.

[05:31] Now, what is this meaningful representation? You will learn it shortly. So, for now, just understand we require a meaningful representation of the data. So, now, if at all if that is your requirement, what kind of model will help us to learn this meaningful representation. We have an auto encoders. So, what is auto encoder? It is an unsupervised deep learning model which primarily focuses on learning the meaningful representation of the data. And why it has to learn a meaningful representation? So that it can reconstruct their own inputs. Whatever it has a learned, it can reconstruct that particular data.

[06:13] So, this is what is the architecture of auto encoder. So, in this particular architecture, we can see there are primarily three components. One is encoder, second we have a latent code, and the third we have an decoder. So, the three important components are encoder, latent code, and then the decoder. Now, what exactly is the working mechanism? We have an input. Input is a passed to to an encoder. Now, when input reaches to an encoder, what encoder has to learn? The What is the role of an encoder? Encoder has to learn the data learn from this data. Now, whatever the encoder is learning, it is a trying to focus on learning the most important features.

[07:04] And towards the end of an encoder, so the output of an encoder is a smaller or an important representation of that input data. So, [snorts] input is passed to an encoder, encoder is learning that data. And what is the output of an encoder? Output of an encoder is an smaller representation of that input data. Which means when I say a smaller representation, what is this information what what what information it consists of? It consists of only those important features of this input. Next, that latent code is passed to to an And what is the role of a decoder here?

[07:45] The role of a decoder is to reconstruct the data back. So, this smaller representation is passed as an input to the decoder, and the decoder will perform some operation, and then it reconstructs the data back. So, input is passed to an encoder. Encoder learns the meaningful features or the representation of the data. So, that the output of the encoder is a smaller internal representation. This smaller internal representation is what is the latent code. We call this as a latent code. And this representation, latent code, is passed to an a decoder.

[08:24] And what is the role of a decoder is to reconstruct the original input. So, this is what is the working mechanism of an autoencoder. The first objective here is to learn that meaningful representation, and the second objective is the decoder has to

reconstruct. Therefore, the two major objectives of autoencoders are first is a representation learning, and second is reconstruction learning. What is a representation learning? Who does this representation learning? The first part of the architecture, where the input is passed to an encoder, and encoder learns the compact representation of the input. So, this is what is the representation learning.

[09:13] That compact representation is what is called representative learning. » [snorts] » Now, this representative learning is passed as an input to the decoder, and the decoder going to will perform a reconstruction. The role of a decoder is to reconstruct the original input. And from where? From this latent representation. So, this is what is the overall architecture, very abstract view. Now, let us understand in detail how this representation learning is going to happen. Now, for example, let us take we have an input.

[09:49] For this particular example, we are Let us consider we are considering an you know, tabulated data. What do you mean by tabulated data? The data which is in the form of rows and columns. Let us assume we have totally 100 features are there. So, here our X is 100 features. So, number of features here. So, that is passed as an input. Now, this is what is the encoder architecture. Now, we have seen sometime back the encoder.

[10:23] This is what is an encoder architecture. So, we will have a different layers there. So, let us now understand what exactly happens when the input is passed to an encoder. Now, let us assume this is the first hidden layer. And in this first hidden layer, there are 75 neurons. So, can you recall from the deep learning concepts how is that we construct a multi-layer perceptron? So, is it not we have an 100 features here? And these 100 features are passed now to 75 neurons. We have a 75 neurons here. Now, how is that it works?

[11:08] Since it is a fully connected neural network, » [laughter] » every of the input is passed to every of the neuron which is connected. So, here since this is a input which are 100 features, and this is the first hidden layer which is having a 75 neurons. So, how is that you will establish the connection? Is it not the first input is connected to all of the neurons? Then the second input is connected to all of these third input and so on. All of these every of the feature is connected to every of the neuron at the hidden layer.

[11:45] First hidden layer. How many such hidden layers are there? There are 75 such there the first hidden layer and how many such neurons? There are 75 neurons in the first hidden layer. So I hope this part is clear to you. This is a simple fully connected neural network. Now, we also seen that what is the mathematical operation that is a performed in the neuron? You will have a summation and you will have an activation function. Similar way, we will have a summation and activation. Summation, activation.

[12:21] So, from this what are we understanding? When we take this summation, the first neuron, this neuron the summation is it not the combination of all of the inputs? For this summation one is reaching, second feature is reaching, third feature is reaching and so on hundredth feature is also reaching. Similar way, for this this is summation is once again the input from first neuron, second, third and so on up to hundred. So this is how the fully connected neural network works.

[12:55] Now, is it not after we perform this operation from here we get one output. From this we get a second output. From here we get the third output and the so on we get the 70 five. So totally because you have placed 75 neurons, 75 output is what we are getting after we performed the first operation. Correct? So the input is 100 that is a passed to the first to layer. In the first hidden layer, we have placed 75 neurons. So So that the output is a 75.

[13:32] Now, my question to all of you here is from 100, when you have reduced to 75, have we deleted any data here? Uh can you think? The answer is no, right? Why? Why we have not? We have not deleted anything. If for this particular neuron, all of the input is being passed. For the second neuron, all of the input is being passed. So now, what is that which has happened? This 100 is being compressed. Right? So we have not deleted any data.

[14:06] We have now obtained the compressed form. So 100 is now reduced to 75. So you need to understand here, we have not deleted from 100 to 75. 25 we have not deleted. This entire 100 is being is being passed to every of the neuron at the hidden layer. Therefore, we have obtained now 75. So that is what is from this part to this part. Now, let us see you have one more hidden layer. Now, in that hidden layer, how many neurons are there?

[14:40] Again, we have 25 neurons. So is it not we have something like this? Now this from 75, you are now reaching to This is a 75. Now you are reaching to 25. Now, how is that this works? Is it not every of the neuron from this is being connected to the next layer next layer neurons? So from here, you are connecting to every of the neuron. From the second, you are connecting to every of these neurons. From third to all of the neurons.

[15:13] Similar way, 75th is connected to all of the neurons. Now, is it not the same operation you are going to perform again? You will have an a summation and you will have an activation. Summation and activation. Is it not that we are going to perform this? Now, if there are 25 neurons, is it not we get the first output, we get the second output, and the similar way we get the 25th output? So, 25 values are what we are going to obtain from this. So, what what what is that we have done?

[15:46] Initially, we had 100. This 100 is a passed to an a first hidden layer which had a 75. Therefore, 100 is reduced to 75. Now, 75 values are now passed to the next hidden layer which is having 25 neurons. Is it not from a 75 you have compressed

down to 25? Does it mean that you have deleted the remaining? Answer is no. Every of the feature is being passed so that now 75 is now compressed down to 25. So, towards the end of this output this layer hidden layer at two, what will be the output? The output is 25. Do you all agree to this? Why 25? It is just for an example to make you understand how you know the encoder works. What is the role of an encoder?

[16:37] To learn the representative learning. What is representative learning? It is a compact representation of an given input. Now, the given input is 100 features. From this, have you not now reduced to 225 which is a compact representation of a given to a 100? Yes. So, the output of this hidden layer H2 is this latent space. So, at the from here to here, what exactly has happened? So, from 75 you have reduced down to 25. So, now the output of the hidden layer H2 is the latent representation. And this latent representation, what exactly has happened? From 100 you have moved it to 75. From 75 you have reduced it down to 25. This 25 is what we call it as an encoded data, or we also call it as an latent

[17:33] code or latent representation. So now, here our features reduced to 25, and let us denote it as H. So, what exactly uh the encoder has a performed? The main role of an encoder here is to map the given input data to a lower dimensional latent representation. And how is that it has achieved? It has achieved by progressively reducing the number of neurons at the hidden layer. So therefore, the dimensions has reduced from 100 to 75, from 75 to 25. Now you can ask the question, is 25 is a fixed?

[18:14] It's an hyperparameter. So you can decide what should be your latent space. Accordingly, you can design the encoder network. Now, this reduction, now from 100 to 75, from 75 to 25, this reduction, is it not has enforced to the encoder to learn the most important features from the input? Because ultimately the aim is to have only 25. If the aim is to have 25, is it not you're forcing the network to learn the most important features only? So that now that the 25 what you have is the compact important representation of a given input. So with us, is it not the first objective of encoder has been achieved? What was the first objective?

[19:02] The first objective is a representation learning. So representation learning is being achieved by an encoder. What is the role? Input pass it to an encoder, make sure you have an uh reduced number of neurons at in the hidden layer. The uh output of the last hidden layer will be the latent representation. Under that latent representation will have a compact representation of a given input, which uh we say that it is a representative learning, which is the first objective of an autoencoder.

[19:34] Now, with this, let us see what exactly happens at a decoder. What is the role of a decoder? The role of a decoder is to reconstruct the data back. Now, let us see how it will reconstruct. This is what we have seen so far. Now, the latent representation we have uh 25 from our previous example, we have seen that the 25

uh number of features or uh features are there at a latent representation. Now, this latent representation is passed as an input to a decoder. Now, decoder is once again a neural network here. And uh since you're working on an uh tabulated data, how at an encoder side you have used a multi-layer perceptron, at a decoder side also you'll be using a multi-layer perceptron uh model only.

[20:23] So, what is the input that you're passing to a decoder? The latent representation is passed as an input to a decoder. Now, let us see again one by one, layer by layer, what exactly is going to happen. Now, initially, is it not this uh 25 uh uh this is the 25. What is this 25? 25 is the input that you're passing to a decoder. At a decoder, we have a hidden layer. So, at this hidden layer, how many neurons are there? So, the this is 25. When it is passed to a first hidden layer at a decoder side, we have 75 neurons. Now, you see there are 75 neurons. This is 25. This is 75.

[21:10] How is the architecture? Is it not every neuron is connected to every other neuron here? So, you will have a connection like this. So, because it is a fully connected neural network, every neuron is connected to every other neuron. Similar way, second neuron is connected to all other neurons in the hidden layer. Similar way, the 25th neuron is connected to all other. So, what exactly is happening here? Now, it is not you have added something. Now, from a 25, what will be the output?

[21:42] Here, you will have again summation activation. Similar way, summation activation and so on, summation activation. Is it not we are going to get now 75 values at the output of this? We are going to have 75 values. Now, from 25, we got to 75. How did we get the 75? Did we add anything extra? The answer is no. Is it not we have expanded now? Yes? From 25, we have expanded. How did we expand in a systematic way? Because every of this neuron is connected to all other neurons. So, which means whatever the output which comes from this is a combination of all those 25.

[22:25] Whatever output which comes from this is a combination of all that 25. Of course, along with that you'll also be multiplying with the weights and then the bias. That is by default here. So, we are now getting 75 output. So, what exactly happened from 25? From this to this, we have expansion. We have what we have performed is an expansion from where? From 25 to 75. So, from 25, we have moved to 75. Now, next we have Uh, from a 75, you are passing it to the next hidden layer. So, how many neurons are there in this hidden layer? There are now 100 neurons are there in this hidden layer.

[23:13] So, now what will be the operation? Is it not every of this is 75? Every of the neuron which is there in this hidden layer is connected to every of the neuron which is connected to the next hidden layer. Answer is yes. So, is it not this? So, every of the input is connected to this neuron. Again, you're going to perform summation and you get an activation, you get the first output. Similar way you do every of the input

is connected to this, you perform summation, activation, you get the second output. So, on so forth, you will now perform summation, activation, you get 100.

[23:51] So, is it not again systematically from 75 you have passed to 100 neurons at the hidden layer. So, from 75, what is that we have obtained? We have obtained 100 values. Is it not that 100 is back your X originally the X value, the original input had 100 features. From 100, is it not we have obtained 100 again? Now, whether that 100 is exactly matching to the original, that we will see later part. But, is it not we have obtained the 100 values back again? So, what exactly has happened?

[24:31] From these 75 values, when you have passed it to an output layer which is having 100 neurons, so we have now obtained the 100 values. So, this is we can say it as a reconstructed output. So, what exactly has happened? I hope you got an intuition behind decoder. Decoder aim is a How is that it is a reconstructing? Now, progressively you're increasing the number of neurons at a hidden layer. And whatever is the output you obtained from the last layer, that is what is the reconstructed input or a reconstructed data, which is same as your original input.

[25:17] So, [snorts] what is the ultimate aim of a decoder? Decoder is not simply recovering the original input by copying the values. We have nowhere copied the values. We have by copying it. We have reconstructed using this compressed information, right? What was the input to a decoder? Is it not the compressed that the 25 latent space that is a passed to a decoder? And then you are progressively expansion. You're doing the expansion.

[25:47] And towards the end we are obtaining the dimension which is the same as your original input. Now, from this what are we understanding? If this latent code or latent representation, if it has a captured the most important features or the structure of the input, is it not the output what we obtain also will be similar to the original input? What do you think about this? The answer is yes. If the latent code is performing well, if the latent code has a good representation, obviously that latent code is passed as an input to the decoder. Definitely the output will be the whatever the output that you are getting will be similar to the original input.

[26:33] » [snorts] » With this, is it not we have obtained the achieved the second objective? What is the second objective? Second objective is a reconstruction learning. With this, is it not we have achieved the reconstruction learning? This is what is the in-detailed architecture of encoder and decoder. So, it has two primary objectives, two components, primary components. One is encoder and decoder. Encoder is primarily used to map the input to an latent representation or a hidden representation H . This hidden representation, we call it as an third component which is called an latent code.

[27:14] The next This latent is being passed to a decoder. What is the role of an decoder? The decoder is has to map the hidden representation H back to the you

know, X . Or this since it is an reconstructed one, we are not calling it as X , we are calling it as an $\hat{X}$. So, this $\hat{X}$ is the reconstructed output. And from where did we get this? We obtained that from H . How did we get that H ? H is obtained from that input. So, [snorts] the primary aim is moving from X to H and from H to reconstructed one $\hat{X}$. Now, so from this, what are we understanding?

[27:58] The entire aim is from here only. First objective is reaching X to H and from H to $\hat{X}$, which means H play a major role here. Correct? Yes. Now, from this architecture, you can understand that why did we take a H value smaller comparing to the input? Is it the dimension of H here is two and which is a definitely smaller than the input? So, is it always compulsory that the latent dimension should have an smaller dimension comparing to an input?

[28:38] What do you think? Can we have an neural network architecture something where the hidden representation is larger comparing to the input. Is this possible? Is both the scenarios possible? Answer is yes. The first one we call it as an under complete. What is under complete? Where the dimension of H is less than the dimension of the input. So, this is important concept. I want everyone to focus here. The dimension of H , that is the latent representation is a smaller than the dimension of the input. So, if we go for this, we call it as an under complete autoencoder. What is under complete? Here the hidden representation will have lesser number of neurons comparing to an or lesser dimension comparing to the input.

[29:33] Now, by doing this, what are we aiming? We are We are ensuring that the or we are enforcing that the model learns the most important features. Now, what is ultimate aim? Ultimate aim is to reconstruct. Now, from where are we reconstructing? We are reconstructing back from this H . Now, H is a lower dimensional one. From this lower dimensional, if you can still able to reconstruct it back, is it not H has captured all the important features?

[30:07] Yes. Is it not H is error you know, loss free encoding of given input? Yes. This is what is an under complete autoencoder. What is under complete? Here, the first objective is from input, you should get that H . From H , you are reconstructing back, which is the second objective. Ultimately, the focus is on this latent representation. Now, what is under complete? Latent representation dimension is a smaller than the original input. If the latent representation dimension is a smaller, which means latent representation aim is to get the most important representation. Next, if size is a smaller, if the dimension is a smaller, ultimately you're reconstructing back.

[30:58] From that a smaller, if you're able to reconstruct back, and if that reconstructed is the same as original, then is it not that H what you have captured has captured the most features, and it is an loss-free encoding of the input. So, this is what is under complete. What is the counterpart? Where the H value is larger than the input. Where the dimension of H , which means you will have a neural network, or

you will have an architecture something like this. Where the H size is the dimension of this will be definitely very much larger than the input. Now, in this, there is one a problem. What is that a problem? Is it not whatever are the six values here, is it not all those six values are copied as it is?

[31:58] Is it not it's a trivial learning? Whatever because aim is from six, you have to get the larger value. One direct way is copy down all those six values to hidden representation as it is. And whatever are the remaining, can we write it as zero? Possible. So, is it not a trivial learning here? What is a trivial learning? It is an identical mapping of the given input as it is to an hidden representation. Since the hidden representation value is dimension is very much greater or equal to an input, the autoencoder would learn a trivial learning. What is a trivial learning? It is directly copying x_i into h and then copying back h to x_i dash. Uh uh so, from x I'm directly copying the values.

[32:50] If you take from four, I'm directly copying the values to six. Uh from four, if I'm copying to six, so remaining two values I'll keep it as zero. Then from six, I'll directly make it to four by removing those values. Where is a learning here? Absolutely, there is no good representation learning. So, therefore, in a real world application or when you're doing a practical hands-on implementation for to solve any of the real world problems, always we will have an architecture which is more uh you know, which is an under complete, where the dimension of an h will be smaller than the given input. So, with this, we have come to an end of this particular session. Since this is the starting of this particular week two, where we have

[33:38] started with auto encoders, our in this particular session, the focus was completely only on understanding what is the role of an auto encoder, what is how encoder works, how decoder works, and the primary two objectives here are representation learning and reconstruction learning. And there are two different types here when it comes to an a latent representation. One is under complete, second is over complete. What is under complete? The dimension of h is less than the dimension of the original input. And what is over complete? The dimension of h is either greater than or equal to an dimension of an input. So, this is all what we have seen in this particular session. In upcoming session, we will see a reconstruction loss. Now,

[34:28] yes, decoder has given the data or output. Now, how far that Now, how far that output is a matching to an original input? So, what exactly is the loss function? So, that we are going to see in the upcoming session. Thank you.